



**Course:** ELEC/COMP 526

**Term:** Fall 2025

**Loc:** HRZ 2010

**Time:** T. Th, 9:25-10:40

## INSTRUCTOR CONTACT INFORMATION

**Instructor:** Peter Varman

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**Office Hours:** T, TH 2:00 – 4:00 pm or by Appointment

## COURSE OBJECTIVES AND LEARNING OUTCOMES

The goal of the course is to provide students with an advanced understanding of the architecture and implementation of modern high-performance computer systems.

Following the course the students will be able to:

- (a) describe the handling of data and control dependencies in high-performance uniprocessors
- (b) design a microarchitecture for pipelined and dynamically scheduled out-of-order processors
- (c) implement simulation models of high-performance pipelined processors
- (d) analyze the behavior of non-blocking and lockup-free caches
- (e) describe the problems of cache coherence in multiprocessor systems
- (f) implement and analyze the behavior of snooping and directory-based cache coherence protocols
- (g) describe the problems of memory ordering and strong and weak consistency in multiprocessors
- (h) describe the implementation of virtual memory and its interaction with the cache hierarchy
- (i) describe architectural support for synchronization including transactional memory
- (j) describe the architecture of modern storage and network systems for datacenter computing
- (k) have familiarity with elements of datacenter computing and system virtualization concepts

## RECOMMENDED TEXTS AND MATERIALS

- Fundamentals of Parallel Multicore Architecture, Yan Solihin
- Parallel Computer Organization and Design, M. Dubois, M. Annavaram, P. Stenström
- Computer Architecture: A Quantitative Approach, (5th edition), J. Hennessy and D. Patterson
- Survey and Research Papers (Will be posted as needed)

## PROJECTS AND EXAM

|                                  |                               |       |
|----------------------------------|-------------------------------|-------|
| Homework Sets and Projects (5-7) | (Assignment-specific weights) | 60-70 |
| Semester Tests                   | (Semester Tests + Final)      | 30-40 |

Projects will involve implementing C programming-based simulation models, running simulation experiments, and analyzing the results.

## LATE POLICIES

All assignments are due by the deadlines.

## RICE HONOR CODE

In this course, all students will be held to the standards of the Rice Honor Code, a code that you pledged to honor when you matriculated at this institution. If you are unfamiliar with the details of this code and how it is administered, you should consult the Honor System Handbook at <http://honor.rice.edu/honor-system-handbook/>. This handbook outlines the University's expectations for the integrity of your academic work, the procedures for resolving alleged violations of those expectations, and the rights and responsibilities of students and faculty members throughout the process.

- All assignments and tests are assumed to represent the individual efforts of a student and are given under the (letter and spirit) of the Rice University Honor Code system.
- Do not consult homework sets, quizzes, solutions or tests from past versions of this class.
- Discussion with other students and teaching staff on assignments is encouraged.
- All submitted work must be done independently by each individual (or lab group if applicable).

**AI-based resources:** Generative AI: Large Language Models (such as ChatGPT and many others) have emerged recently. The goal of this course is for students to be able to create their own code to develop unique implementations of micro-architectural structures. As such, generative AI methods are not to be used in this course.

## MENTAL HEALTH STATEMENT

The wellbeing and mental health of students is important; if you are having trouble completing your coursework, please reach out to the Wellbeing and Counseling Center. Rice University provides cost-free mental health services through the Wellbeing and Counseling Center to help you manage personal challenges that threaten your personal or academic well-being.

If you believe you are experiencing unusual amounts of stress, sadness, or anxiety, the Student Wellbeing Office or the Rice Counseling Center may be able to assist you. The Wellbeing and Counseling Center is located in the Gibbs Wellness Center and can be reached at 713-348-3311 (available 24/7)."

## DISABILITY SUPPORT SERVICES

If you have a documented disability or other condition that may affect academic performance you should: 1) make sure this documentation is on file with Disability Support Services (Allen Center, Room 111 / [adarice@rice.edu](mailto:adarice@rice.edu) / x5841) to determine the accommodations you need; and 2) talk with me to discuss your accommodation needs.

## SYLLABUS CHANGE POLICY

This syllabus is only a guide for the course and is subject to change with advance notice.

## Topics

### 1. Processor

- Instruction Level Parallelism
  - Simple static pipelining
  - Dynamically-scheduled pipeline processors
  - Superscalar, VLIW
- Data Parallelism
  - SIMD, Vector and Stream Processors, GPGPUs
- Thread Level Parallelism
  - Multithreaded Processors
  - Multi Cores
  - NUMA and multiprocessor processors
  - Clusters
- Domain-specific Accelerators
  - GPU, TPU, SDN, Compute-In-Storage, others

### 2. Memory

- Uniprocessor Caches
  - Organizations and Addressing
  - Lockup Free Caches
  - Multi-level Cache hierarchy
- Multiprocessor Caches
  - Cache Coherence
  - Snooping and Directory Protocols
  - MSI, MSEI, MSEOI, MESIF protocols
- Memory Orderings and Memory Consistency Models
- Virtual Memory
  - Demand Paging
  - TLB and Page Walks
  - Processor Caches and Virtual memory
  - Virtual and Physical Caches

- Emerging Memory Devices
  - NVM, HBM, PIM, others

### **3. Synchronization**

- Spin Locks
- Sleep Locks
- Lock-Free Synchronization
- Transactional Memory

### **4. I/O and Storage**

- Bandwidth, Latency, Topology
- Scalable Networks : Clos, Butterfly, Fat Trees
- PCIe, CXL, RDMA (InfiniBand, RoCE)
- Distributed storage architectures: Reliability, RAID, Erasure Codes

### **5. System Resource Virtualization (Software Defined Everything (SDX))**

- Sharing vs Isolation
- Resource Virtualization Techniques
- Resource Scheduling